Haocheng He

Phone: +852 5698 7112 | haocheng.he@foxmail.com | Hong Kong SAR, China
linkedin.com/in/haocheng-he-92657028b

Professional Summary

Aviation Engineering MSc candidate at PolyU (GPA: 3.95/4.3), with an IE undergraduate and Law minor from GDUT. Experience spans airport simulation (AnyLogic, 5 routing algorithms), manufacturing optimization (line balance 58%→70%), and Ascend 310B hardware R&D. 3 patent applications.

Education

The Hong Kong Polytechnic University — Hong Kong SAR, China
Master of Science in Aviation Engineering and Operations Management — Sep 2025 – Jan 2027
GPA: 3.95 / 4.3 (Current)
Thesis: Simulation Modelling and Optimization for Airport Ferry Vehicle Scheduling under Uncertainty

Guangdong University of Technology — Guangzhou, China
Bachelor of Engineering in Industrial Engineering — Sep 2021 – Jun 2025
GPA: 3.67 (Top 20%)
Thesis: AnyLogic-Based Simulation Optimization of Emergency Vehicle Route Planning in Large Airports
Honors: University-level Excellent Graduation Thesis, Graduation Design Innovation Award
Bachelor of Laws (Minor Degree in Law) — Sep 2022 – Jun 2025
GPA: 3.61
Thesis: Coordination Between Copyright Protection and Privacy Protection in Digital Rights Management

Experience

Assistant AI Engineer — May 2025 – Jun 2026
Shenzhen Wuxian Hongyin Tech. — *Shenzhen, China*
- Participated in Ascend 310B smart piano R&D across 3 development phases; tested 10+ boards (CPLD, STM32), iterated motor driver 3 times, developed firmware and authored 5 technical documents.
- Participated in camera- and Ascend-based piano fingering recognition module integration (Python/C++); diagnosed cross-layer issues.
- Maintained Ubuntu servers, configured Huawei network equipment, operated 3D printer, supported technical interviews.

Research Assistant — Sep 2023 – Jun 2025
GDUT (with Guangzhou Baiyun Int'l Airport) — *Guangzhou, China*
- Built an airport road-network, apron, and runway-area simulation model in AnyLogic, Java, and MySQL for aircraft landing and emergency fire-truck dispatch scenarios.
- Designed single/multi-vehicle, single/multi-event, and runway-road-closure scenarios; compared D* Lite, A*, Dijkstra, BFS, and GA routing algorithms.
- Completed 4 emergency-scenario comparisons across 5 routing algorithms; D* Lite achieved the highest average reliability index (1.342).

Industrial Engineering Intern — Apr 2024 – May 2024
Midea Group — Kitchen & Water Heater Division — *Foshan, China*
- Conducted stopwatch time studies across all workstations on a dishwasher sheet-metal line, built operator workload tables, and identified 5 over-cycle steps.
- Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%, reducing headcount from 20 to 18.

Manufacturing Operations Intern — Jul 2023 – Aug 2023
Guangxi LiuGong Machinery Co., Ltd. — *Liuzhou, China*
- Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis.
- Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation.

Projects

Aircraft Engine Remaining Useful Life Prediction (Course Project) — Apr 2024 – Jul 2024
Guangdong University of Technology — *Guangzhou*
- Built an RUL prediction workflow on NASA C-MAPSS FD001 multivariate time-series data; cleaned operating-condition data and 21 sensor signals, constructed sequence features, and applied PCA.
- Compared LSTM, BiLSTM, Transformer, and Random Forest baselines for time-series forecasting; after iterative tuning, the BiLSTM model reached about 8% final error from an initial model error of about 50%.

Skills & Intellectual Property

Programming: Python, Java, SQL/MySQL, C (Embedded/STM32), JS/TS
ML & Algorithms: LSTM, BiLSTM, Transformer, Random Forest, PCA, Dijkstra, A*, BFS, GA, OR-Tools
Engineering: AnyLogic, Plant Simulation, Ubuntu/Linux, Git, JLC EDA, Oscilloscope, Logic Analyzer
AI: LLM Workflows, Prompt Engineering
Languages: English (IELTS 6.5), Mandarin (native), Cantonese (conversational) **Patent Applications:** *MCMC Motion Control for PI Coating Machine Controller* (CN122085794A, 2026); *Coating Control via Deep Reinforcement Learning* (CN121325552A, 2025); *Energy Regulation for Household Solar Storage* (CN117767437A, 2024). UM: *Wire Harness Production Equipment* (CN221352455U, 2024). SW: Financial Data Analysis (2024SR0146827); Flood Dispatch (2024SR0549247).